

# Gonzalo López Crespo

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## About Me

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AI and Python developer with a solid mathematical foundation, focused on building generative-AI solutions for real use cases: RAG systems, LLM-based agents and the integration of AI APIs and models into working applications. I combine a strong Python core (business logic, services and integration) with experience in data preparation and transformation and sound engineering practices (Git, testing, reproducible pipelines), and I am comfortable iterating on prototypes until they reach production alongside technical teams.

## Technical Skills

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- **Generative AI & LLMs:** RAG system design with embeddings, semantic search and vector databases, contextual-memory management and prompt engineering; LLM-based agents and chatbot architectures adaptable to business environments and hardened against prompt-injection; integration of AI APIs and models into real applications.
- **Python & service development:** pandas, numpy, asyncio, aiohttp, FastAPI (REST), Selenium, BeautifulSoup and Pytest. Business logic, API integration and consumption, low-latency asynchronous services, large-scale web scraping, scripting and automation.
- **Machine Learning & Deep Learning:** scikit-learn, PyTorch, PyTorch Geometric. Regression, Random Forest, clustering and Bayesian methods; dense networks, CNN, LSTM, Graph Neural Networks (GCN, GAT) and spatio-temporal modelling; evaluation (MAE/RMSE/R<sup>2</sup>), model selection and experiment tracking with MLflow; NLP and text processing.
- **Cloud & infrastructure:** Azure Data Factory, Blob Storage, SQL Azure, Cosmos DB and Azure AD; App Services, Azure Functions and containers (Docker) for deploying scalable solutions; initial exposure to AWS, BigQuery and Hadoop. Training oriented to the AZ-204 certification.
- **Data preparation & BI:** Databricks and PySpark at scale; ETL and Medallion architecture (Bronze/Silver/Gold) with Azure Data Factory and Pentaho (PDI); Microsoft Fabric (Lakehouse, OneLake, Dataflows Gen2); SQL and T-SQL, relational design and normalisation; Power BI dashboards (star schemas, DAX, Power Query).
- **Engineering & tooling:** Git and peer code review; systems programming in C (manual memory and process management, 42 Málaga); shell scripting and automation on Unix/Linux.
- **Maths & statistics:** probability, linear algebra, optimisation and inference (including A/B testing) as a direct foundation for modelling and AI systems.

## Projects

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### MálagaHelper — UrbanMind (AI city assistant) · Hackathon project

Conversational urban-intelligence assistant for Málaga built at a hackathon.

**Demo:** [malaga-helper.vercel.app](https://malaga-helper.vercel.app)

- **LLM integration:** adaptable conversational assistant with real-time multilingual UI translation via an LLM (Llama 3.1 through OpenRouter), integrating the model through its API.
- **Data & product:** ingestion of crowd-level, museum and monument datasets, saturation computation, a 2-hour forecast and an interactive map panel; web stack Next.js, React and TypeScript.

### Flight-delay propagation prediction in air-traffic networks · MSc Thesis · ESESA

Modelling the US airport network as a directed graph to predict how delays propagate using spatio-temporal neural networks.

**Repository:** [github.com/gonzalona/flight-delay-propagation](https://github.com/gonzalona/flight-delay-propagation)

- **GNN architecture:** progressive design from dense networks and LSTM up to GCN and GAT, culminating in a spatio-temporal model combining GNN with LSTM to capture structural and temporal dependencies.
- **Data pipeline:** Kaggle dataset of ~70 M records (2018–2022) with a modular architecture for preprocessing, multi-horizon training and evaluation; unit tests (Pytest), hyperparameters via YAML and Git.

## Digital-asset market automation & analysis system

An end-to-end, self-directed Python ecosystem for real-time analysis and algorithmic investment in secondary digital markets (Steam).

- **Software engineering (Python async):** open-source extension of the steampy library to support asynchronous requests (asyncio, aiohttp), critically reducing order-execution latency; 24/7 availability with rotating proxies and session management.
- **ETL & data:** real-time extraction and cleaning scripts (Selenium, BeautifulSoup) consolidated into own databases; financial-control dashboard in Power BI on a dimensional model with DAX measures.

## Education

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**MSc in Big Data & Artificial Intelligence** · ESESA · In progress

- Generative AI and RAG with embeddings, semantic search, contextual memory and prompt engineering; NLP and language models; ML/DL in Python (regression, Random Forest, dense networks, CNN, LSTM).
- Large-scale data processing with Databricks and PySpark; end-to-end projects with Microsoft Fabric, Azure Data Factory and Medallion architecture; Power BI dashboards.

**BSc in Mathematics** · UNED · In progress

- Third-year student. Solid foundation in probability, statistics, linear algebra and quantitative reasoning, applied to modelling and the development of AI models.

**42 Málaga student** · Fundación Telefónica · Nov 2023 – Nov 2025

- Systems programming in C (Minishell with manual memory and process management); algorithms, shell scripting and automation in Unix under peer-review methodology.

## Certifications & Courses

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**Microsoft Certified: Power BI Data Analyst Associate (PL-300)** · Certification

**IFCD21 — Developing Microsoft Azure Solutions (AZ-204)** · 100 h · Arelance

**Applied Machine Learning with Python** · 150 h

**Master in SQL Server** · 40 h · IBM SkillsBuild

**Other courses** · Data Analysis with Power BI Desktop (40 h) · Data Analytics (50 h) · Ethical hacking: expert (120 h)

## Languages

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- **Spanish:** native
- **English:** very high level (C1)